qualityEvaluation

Service Description

Service ID: *"qualityEvaluation"*

Abstract
This document describes the `qualityEvaluation` service, version 1.0.0, produced by the DeviceQoSEvaluator support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `qualityEvaluation` service, which provides quality-of-service evaluation of candidate providers from measurements of the devices hosting them (round-trip time, CPU load, memory use, network load). It is produced by the DeviceQoSEvaluator system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

New in Arrowhead 5; Arrowhead 4.x had QoS monitoring inside the Orchestrator.

2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration calls `filter` or `sort` with the candidate providers and the consumer's QoS requirement (`enable.qos=true`). The evaluator combines the requested metrics with their weights over the time window and filters by the threshold or sorts by the score.

2.3 Important Delimitations and Dependencies

Only providers whose devices are measured can be evaluated; others are reported in the warnings. Measurements are taken by the evaluator (RTT) and device agents (augmented metrics).

3 Service Interface

This section describes the interfaces to the `qualityEvaluation` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

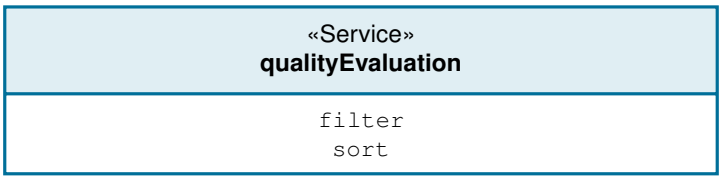


Figure 1: SysML-style block of the `qualityEvaluation` service and its operations.

The following interface operations are available.

3.1 operation filter (QoSEvaluationRequest) : QoSEvaluationFilterResponse

Returns the providers that meet the threshold of the weighted metrics, and the dropped ones.

3.2 operation sort (QoSEvaluationRequest) : QoSEvaluationSortResponse

Returns the providers ordered by the weighted metrics, best first.

4 Information Model

Here, all data objects that can be part of the `qualityEvaluation` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.4, which are used to represent things like hashes and identifiers.

4.1 struct QoSEvaluationFilterResponse

Result of a filter evaluation.

Field	Type	Description
passedProviders	List<String>	Providers that meet the threshold.
droppedProviders	List<String>	Providers that do not meet it.
warnings	Map<String, List<String>>	Per provider, warnings such as missing measurements.

4.2 struct QoSEvaluationRequest

Providers to evaluate and the evaluation settings.

Field	Type	Description
providers	List<Name>	Candidate provider systems.
configuration	Map<String, Object>	Evaluation settings: `metricNames` (e.g. `RTT_MEAN`), `metricWeights`, `timeWindow` (seconds) and `threshold`.

4.3 struct QoSEvaluationSortResponse

Result of a sort evaluation.

Field	Type	Description
sortedProviders	List<String>	Providers ordered from best to worst.
warnings	Map<String, List<String>>	Per provider, warnings such as missing measurements.

4.4 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Object	Any structured value.
String	A chain of characters.

5 References

[1] J. Delsing, “DeviceQoSEvaluator – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-DeviceQoSEvaluator-java-spring/documentation/v5.3.0/DeviceQoSEvaluator-v5.3.0-SysD.pdf.

[2] ——, “qualityEvaluation HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.3.0, 2026, deviceQoSEvaluator-v5.3.0-qualityEvaluation-HTTP-IDD.pdf.

[3] ——, “qualityEvaluation MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.3.0, 2026, deviceQoSEvaluator-v5.3.0-qualityEvaluation-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	